

"It's a Feeling That You're on The Top": Embodied Virtual Reality Metaphors to Promote Achievement in a Collaborative Language Learning Experience

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Abstract

Adult language learners encounter emotional challenges such as foreign language anxiety that can affect their ability to communicate or progress in building their language skills. In this work we explore how the design of spaces for language practice can promote positive affective responses to promote a sense of achievement and the impact of Virtual Reality immersion and environments in facilitating this. The authors design a fully immersive VR experience with interactive metaphors for growth and achievement in which users navigate a campsite and collaborate on a storytelling quest to build their language skills. The design is evaluated with 11 participants. Qualitative analysis of semi-structured interviews revealed several key patterns in participants' experience, including positive emotional reactions and feelings of achievement following the activity.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Collaborative interaction*; **Empirical studies in HCI**.

Keywords

Language Learning, Virtual Reality, Immersion

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1 Introduction

Language learning can be deeply impacted by a learner's positive and negative emotions and beliefs, and particularly in settings where the learner is surrounded by their target language, foreign language anxiety can occur [32, 45]. Adult language learners in particular, who may be starting over in a new context can find themselves comparing themselves to peers who are proficient in the target language, face fear of judgment by other adults, and self-esteem challenges as they navigate settlement. These can culminate in sentiments of hopelessness and fixed mindset about their language journey, instilling feelings of low self-efficacy, helplessness, and performance avoidance [1, 5, 13, 20, 36, 37, 43, 54]. Proficiency in speaking requires that learners achieve discourse competence, which focuses on both comprehension and production of speech, and is a challenge for adult learners particularly, who across varying level of proficiency, can attain grammatical competence – be able to comprehend vocabulary and sentence structures – but struggle with discourse [46]. Compared to young language learners who may have structured resources like English as a Second Language (ESL) curriculum and peer networks through their schools to practice speaking, research into the needs of adult immigrants and language learners has revealed there is a necessity for more resources that provide formal or informal opportunities for language acquisition, as spaces that cater to these needs face resource limitations, and current digital means such as mobile apps are limited in the speaking opportunities they provide learners [1, 49].

Collaborative games are one avenue through which learners can experience a sense of achievement, which can yield positive affect. Furthermore, collaborative games can embed socialization to facilitate peer networks and support. In recent years, the use of social Virtual Reality (VR) for social interaction [22, 52] and providing mental health support [2, 3] suggests an opportunity to facilitate community-building, empowerment, and group affirmation. Through this work we seek to understand how the intersection of learning and socialization can be intertwined to benefit adult language learners in cultivating spaces for practice which may be not accessible in their physical locality. Compared to other mediums,

VR offers a variety of affordances including ambient and audio capabilities and presence and engagement [19, 21], which contribute to more immersed experiences that can lead to vivid and connected learning experiences. Although researchers and VR designers have explored the potential for immersive games and activities to impact learning, there is limited research focusing on the ways in which VR can be designed to counteract negative affect related to adult language learning, thus we ask the following questions in this work:

- To what extent can a virtual experience designed for embodied visual metaphors and collaboration promote a sense of self-efficacy in adult language learners' attitudes about their language journey?
- How do virtual affordances such as synchronous communication, interactivity, co-presence, and audio impact the experience?

In approaching these questions, we emphasize social and achievement-based gamer motivation principles in the design and the development of a learner-mediated virtual experience [53] and utilize visual metaphors of growth to embed the experience with empowering elements.

2 Related Works

2.1 Collaborative and Game-Based Learning

Collaborative games have been explored throughout HCI for their role in promoting socialization, fostering digital communities, and increasing engagement in learning [55] by adopting game theories such as player motivation [42, 53], and engagement theories [47]. In doing so, collaborative games can promote team-building [15] enrich understanding of complex situations, and foster empathy and inclusion across players from different backgrounds [44]. Multimedia learning principles [35, 38] introduced by Mayer, are a set of principles revealing how incorporating multiple modes of information can benefit learning experiences. A number of studies in the learning sciences have investigated the use of multimedia learning principles: using redundancy principle with text and callouts in learning videos to improve learners' comprehension [6], spatial contiguity principle to place text together, resulting in learner's incidental learning [40], and modality principle through combining spoken words and visuals across a number of interactive games to enable deeper learning [34, 39].

2.2 Experiential Language Learning in Virtual Reality

VR has emerged as a collaborative space, but in the language learning context has predominantly focused on vocabulary building through single-player experiences. VR Experiences such as learning Spanish vocabulary for household items, [17] which focused on single player and non-dialogue based learning, revealed to be effective for context-based language learning. Similarly, learners navigating a virtual apartment with connected text-based vocabulary had higher retention of Swedish words compared to traditional flashcards [14]. Although they do not explore social and psychological components of the language learning experience, both of these studies convey the cognitive potential of immersion for language learners. Similarly, ESL college students using VR for individual

learning tasks in VR before engaging in free-form socialization with English speakers across the world [29] had positive perceptions of their embodiments in VR environment [29] and Cheng et al explored teaching language and culture with dialogue in VR using head-mounted device and NPC avatars [9].

2.3 Learners' Affect in Virtual Reality

Self-Determination theory explains the extent to which a person's psychological and social context play a key role in their motivation [27, 42], indicating that the design of a learner's environment can play a part in their motivation and achievement. As such, research in virtual reality has explored the impact of different virtual environments on users. Open, nature spaces, for example, have been shown to induce feelings of awe in virtual users and promote positive affect [10, 23]. The Proteus Effect, defines how a user's behavior changes as a result of being embodied through avatars. HCI research has demonstrated that perception-action enabled experiences in VR are conducive to behavioral and brain outcomes for learners [28], and that VR experiences can inspire students' curiosity and promote continued engagement with a task [33]. However, a limited set of research has observed the affective impact of VR experiences on language learners. We find two studies that looked at adjacent effects: in the first, learners' oral performance and learning motivation were enhanced by doing cooperative task-based learning in VR [26], and in the second, French learners' anxiety and oral comprehensibility was positively impacted after being immersed in a virtual social experience [50]. These findings indicate the potential for VR to serve as a space for social language learning experiences that positively impact learners' affect. Thus, we build upon these findings to present the design of a dedicated virtual activity for language learners and contribute preliminary findings at the intersection of social VR language practice and affect-focused design.

3 Design

We implemented a multiplayer storytelling experience in META Horizon Worlds, mediated through the META Oculus 2 Virtual Reality headset and controllers. We designed immersive features to support language learners on a virtual quest to learn and interact with new words and use these words in a collaborative storytelling activity using three embodied metaphors of growth. The virtual experience accommodates synchronous engagement of groups of players with the following goals:

- (1) Incorporate Embodied Metaphors of Achievement: Design virtual features that enable users to visualize growth as they interact [53]
- (2) Socialization: Design a collaborative experience that promotes opportunities for peer-support and cooperative learning
- (3) Facilitate a Self-directed Learning Experience: Design Interactive features that gives users agency in their learning journey [16].

Research into adult language learner speaking has found that learner-learner interactions can help promote discourse competence

[46]. Collaborative storytelling is one activity used in conversational classrooms that incorporates both comprehension and production, as it is simultaneously an oral and interactive activity. We designed this experience to provide scaffolding for learning vocabulary words at the start before the speaking stage, which would encourage speaking through turning-taking. Learners at all proficiency levels can face the challenge of discourse competence even if they have achieved grammatical competence. Thus, although the words selected for this activity were aimed at intermediate language learners, players of other levels could still meaningfully engage in utilizing or practicing their speaking once they gained familiarity with the words through the embedded learning scaffolding.

3.1 User Experience

Users are immersed in a camp environment with a forest they must traverse, collecting interactive vocabulary words, or VocabLogs that produce the pronunciation of words. We design a word exploration and learning area, or Mushroom Field, a space in which learners can hear sentences using the VocabLogs they have collected. After users have discovered, learned about, and collected all their words, they gather at a campsite. Users engage in turn-taking, facilitated through numbered VocabLogs, to create a collaborative story using their vocabulary words. After each new part of the story, users drop the VocabLogs into the firepit, signifying their contribution to the growth of the story. The campfire is designed to progressively grow in flames, size, and color as more words are added to the firepit. When the last player has added the last part of the story, a book spawns. Participants then carry this book to the top of an interactive staircase that emerges when they grab the book. Once the book is placed on a bookcase, confetti fills the environment.

3.2 Interactive Self-Directed Learning Features

VocabLogs We adopt the redundancy principle in incorporating both audio and visual channels of understanding the vocabulary words that users collect [51]. VocabLogs have an attached mushroom which when touched by a user pronounces the vocabulary word as audio, shown in Figure 1.

Mushroom Field We adopt the Contiguity Principle [12] in designing the Mushroom field, shown in Figure 2, a space in which users can gain deeper practical understanding of their words. Touching a mushroom of the corresponding color in the field enables participants to hear an example sentence of the word, helping them understand and analyze the meaning of the word based on the context.

3.3 Embodied Metaphors

3.3.1 CampFire: The campfire pit, shown in Figure 3, starts as empty and grows with flames of increasing size as participants add VocabLogs inside. Participants receive haptic feedback when a new VocabLog is added. Each log contributes to the growth of a communal fire, encouraging feelings of collective agency, mutual support, and a sense of shared purpose.

3.3.2 Storybook: At the completion of the collaborative story, a physical book materializes. This book serves as a tangible representation of the learners' collaborative efforts, with a goal of instilling a sense of accomplishment [25]. Having tangible representations of

efforts can promote feelings of ownership and accomplishment in players, thus we incorporate this interactive storybook that appears when the Endlog is thrown in the Fire Pit [4, 7, 18]. Once players pick up the book, the staircase emerges.

3.3.3 Completion staircase: To promote an embodied metaphor of being uplifted, we design the staircase to move users up towards the sky. The staircase, shown in Figure 4, increases by 3 steps as users climb, elevating them above the forest and campfire scene, associated with progression and power [11]. By altering their visual perspective after facing a challenge and "rising above" language difficulties, we anticipated a mood of empowerment and confidence in their ability to succeed, leading Learners to evaluate their participation and abilities positively [11, 31]

3.4 Environmental Design

We design the virtual environment with vast open skies that reflect a starry night under which is a woody camp surrounded by mountains. Simulated natural environments can promote learning while reducing negative affect by providing a "calmer, quieter, safer" context for learning and encouraging more prosocial behavior [24]. The natural landscapes, with towering trees above the avatar's height, a dark forest, and the starry night sky, are all intentionally designed to induce a feeling of awe, which can trigger players to show more generosity or help toward others during the experience [10, 41]. We designed the current experience after a first iteration piloted with 6 English Language Learning university students, whose insights informed our incorporation of a tutorial room, audio-based learning scaffolding, and haptic feedback for further immersion.

4 Evaluation

We conduct this initial evaluation by recruiting English language-learning adults in the local area through flyers and in collaboration with organizations providing language-learning resources. We have enrolled a cohort of 11 participants (10 females and 1 male, with ages ranging from 32 - 66, with a mean age of 41 years old), detailed in Table 1. Study activities took around 90 minutes in total, with the virtual experience taking 25 minutes, and interviews and survey completion taking approximately 45 minutes. Participation was voluntary and participants were compensated a total of \$25 USD. For the purpose of this study, participants had a basic conversational level in English as interviews were conducted in English following a similar convention as other research in the domain to reduce translation cost [8, 30]. We aimed to represent intermediate English learners but made the study open to participants across all levels, as producing speech has been shown to be a challenge across all-levels. For 1 session which included a beginner and lower-intermediate learner, the researcher was available to provide an additional sentence or alternative definition for the words in the room when requested. Before the activity, users experience a tutorial room to become accustomed to actions needed for the experience. Users participated in the virtual experience in groups of 2 and 4. We conducted semi-structured group interviews with questions about social interactions in the VR space, impact of VR storytelling on language learning, emotional states throughout the activity, evaluation of virtual features and design, considerations

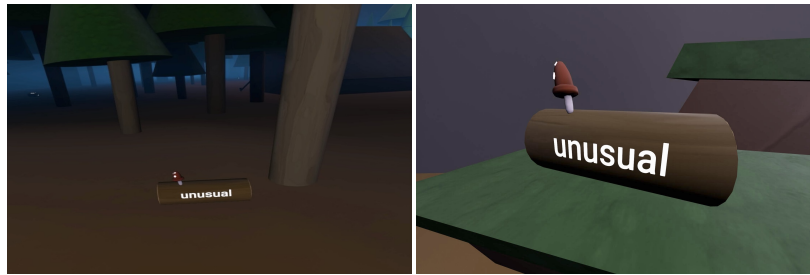


Figure 1: VocabLogs and accompanying Sound Mushrooms

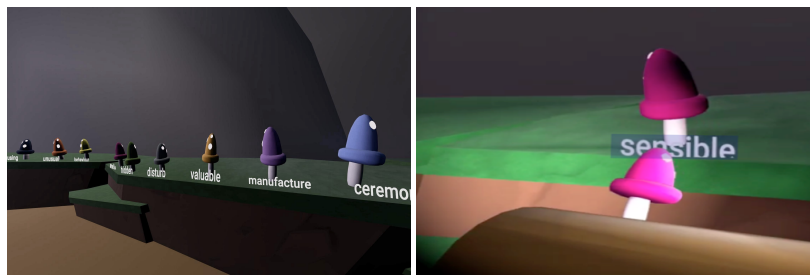


Figure 2: The Mushroom Field



Figure 3: The transformation of the Fire initiated by the VocabLogs

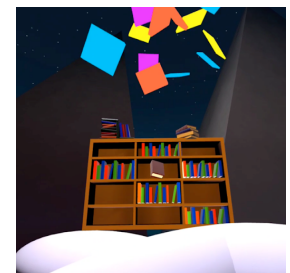
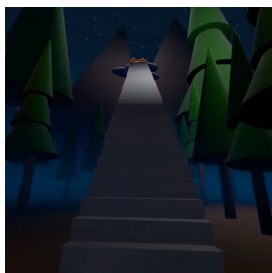


Figure 4: Stairs and Bookshelf

for potential changes, and reflections on motivation and interest in continuing language learning within the VR environment. All parts of the protocol were approved by the Institutional Review Board of the researcher's institution. The qualitative data recorded from interviews was analyzed thematically through a grounded theory approach [48]. Researchers independently analyzed the same

set of recordings in all sessions using open coding; each listening to the recordings alongside transcriptions to break them into parts. Each researcher then organized and categorized their codes broadly using a Miro digital collaborative whiteboard tool. Finally, each researcher performed selective coding, grouping categories,

Table 1: Demographics of Participants

ID	Gender	Age	Home Country	Language Level	Duration in the U.S.
P1	M	66	Spain	Advanced	26 years
P2	F	62	Spain	Advanced	50 years
P3	F	36	China	Upper Intermediate	1 year
P4	F	32	China	Intermediate	5 months
P5	F	55	China	Intermediate	5 months
P6	F	32	Russia	Beginner	11 months
P7	F	34	Russia	Lower Intermediate	2 years
P8	F	33	Brazil	Intermediate	1 year
P9	F	35	Korea	Lower Intermediate	16 months
P10	F	37	Korea	Upper Intermediate	18 months
P11	F	34	France	Intermediate	1 Year

and connecting ideas to name overarching themes developed. Researchers then brought these themes into the discussion to compare the overlapping and recurring themes across all analysis sessions.

5 Preliminary Findings

In this study, we designed a virtual experience. We detail our preliminary findings from the sessions and analysis of our interviews below.

5.1 Embodied Metaphors and Environment

Our first research question asked: To what extent can a virtual experience designed for embodied visual metaphors and collaboration promote a sense of self-efficacy in adult language learners' attitudes about their language journey?

5.1.1 Embodied Metaphors contributed to a strong sense of achievement after completing the activity. We found that participants experienced a strong sense of achievement after completing the activity. Participants developed a mental association of growth with the fire; P2 stating "I saw it, the flames got higher as you put more logs, it grew (P2)" The visual growth of the fire motivated them to contribute to the story, P1 noting "...I'm gonna put more (P1)." The emergence of the book induced feelings of wonder and materialized their collective achievement. P11 reflected that "it was a symbol of our all-together story (P11)." Using the words to complete the activity prompted a sense of confidence in user's motivation to use the new words in real-world situations: "if I use it in the story, maybe I will use it in the real life. It's more confident than before (P4)". After completion of the task, participants noted feeling very "proud (P9)" and "confident" (P4, P10). The elevated view from the top of the stairs reinforced a strong sense of achievement—"it's a feeling that you're on the top at the same time that you did your task (P7)." Participants expressed that the elevated view was "beautiful (P11)," as if "the world under your feet (P7)," and that "when I turned around and saw how high I was... it was very rewarding (P1)." Participants described the overall experience as igniting a sense of excitement and "adventure" (P5) The conclusion of the experience left participants with feelings of "triumph"(P7) and "freedom"(P11). These insights reveal that **integration of these visual metaphors yielded positive affective responses in participants, as the**

immersion and embodied actions had a positive influence on learners in the virtual experience.

Our second research question asked: How do virtual affordances such as synchronous communication, interactivity, co-presence, and audio impact the experience? Our insights demonstrate these affordances played a role in promoting socialization, collaborative learning, and strong visual and sensory associations, which we detail below.

5.2 Socialization in the Virtual Experience

5.2.1 Language Learners favored Group and Paired Activities for Socialization Opportunities. The majority of participants reflected positively on the social aspect of the activity. Participants viewed the group-based nature of the activity as a socialization opportunity to facilitate community building and the expansion of their social networks online. For P9, this was notable because "it's really hard to meet a person (P9)" and P8 suggested "maybe you can meet new people in a game like this (P8)", while P5 reflected on the usefulness of an activity like this in cases where an individual had limited social circle: "if in some situation where you don't have friends around, you can also practice at home (P5)." We see this comfort in socialization reflected in participant's high rating of the their comfort in interacting with others in the virtual space, shown in Figure 5, question 2. Participants (P1, P4, P7, P8) also found the activity being completed with others to be more favorable than a solitary experience: "I think it's better than being, like, alone (P8)." However, there was a desire for more interaction throughout other parts of the experience: "Seems like we just do by ourselves, we do not have much interact (P5)." As a result of this, participants indicated that more opportunities for engagement in the virtual experience would be favorable, and facilitating this by having users actively collaborate at different stages of the experience beyond the firepit could enable this. P4 described wanting this to be embedded more explicitly into the game: "we can do warm up game before the storytelling to be familiar with each other (P4)" while others viewed it as something that should be prompted before the activity: "if we had spoken only one hour before, it would have been more comfortable (P11)"; this is also reflected in Figure 5, question 2 regarding some players discomfort interacting with others in the virtual space. **The VR environment was seen as a valuable tool**

for social engagement, particularly for those with limited opportunities for real-world socialization, which could be further enriched through activities that foster pair-bonding.

5.2.2 Collaborative Storytelling Prompted Presence and Language Production. Beyond the acquisition of vocabulary, participants found the collocated storytelling prompted by the Campfire interaction to be an effective collaborative learning experience. The story’s interconnectedness and the turn-taking prompted by the sequential VocabLogs prompted them to be present and attentive during the experience: “It’s like group working because we need to concentrate on sentences that were before you (P7)” and “other people can give you some idea, you can follow the line of the story (P5).” They also acknowledged that paying attention to other participants enhanced their learning process by stimulating new ideas and creativity: “I really liked connecting to other people’s sentences because I can make my imagination (P10).” Participants experienced a boost in confidence about their ability to use words they learned in the experience: “if I use it in the story, maybe I will use it in real life. It’s more confident than before (P4).”

5.3 Designing for Self-Guided Interactive Learning

5.3.1 Interactive Learning Features Promoted Empowerment, Contextual Learning, and Peer-Support. Participants evaluated the storytelling activity around the Campfire using the VocabLogs as a tool for understanding how words were used contextually (P3, P8) and particularly highlighted the ease of remembering words by using them in sentences: “a lot more effective than just reading or something using a screen kind of passively (P2).” Moreover, participants viewed learning with others as an opportunity to help others, build together, and strengthen their own useage of language regardless of their language levels. Advanced participants viewed it as an opportunity to use words they already knew and generate new synonyms: “I’m also trying to find the new synonyms for this words ... it also was helpful for me (P7).”

5.3.2 Immersive Multi-media through Audio and Visual Cues Promoted Vivid Associations. Participants perceived the fully immersive nature of the experience as contributing to their retention and learning: “My body helps my mind to work together (P6)” and “I think it is such an intense experience that I am going to remember it very vividly because it incorporates so many dimensions (P2).” This suggests that the multimedia aspects of the VR experience could ingrain strong sensory impact in users which could enhance their retention. Relatedly roughly 91 percent of participants reported a high level of immersion (greater than 7 on the Likert scale) in the VR environment as shown in Figure 5, question 4. In summary, participants revealed the potential for the multi-media immersive environments to improve their language learning experiences by enhancing memory retention and boosting learner confidence which facilitated peer-support.

5.4 Challenges to the Embodied Experience

Our findings also revealed negative aspects of players’ experiences that should be addressed to make the experience more effective, enjoyable, and conducive to their language learning needs. First,

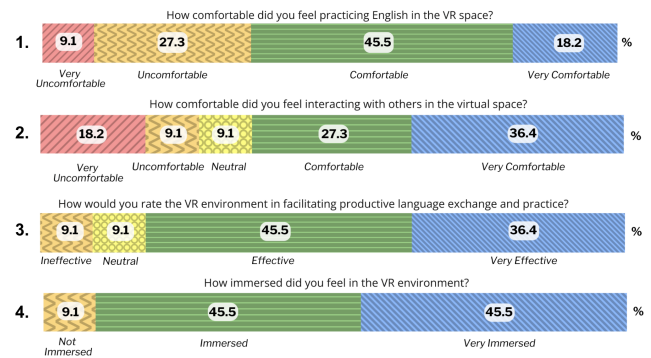


Figure 5: Four Horizontal Stacked Bar Charts with Participant Responses to VR Experience

in regards to calibrating language levels, participants revealed that the opportunity to progress between levels would be beneficial: “if there is an easy level, the next level would be a little bit harder (P8).” One beginner participant, P6, was included in this session who revealed the word choices were indeed too difficult for them, which may have contributed to a negative experience practicing speaking, reflected in Figure 5, question 1. Although we welcomed diverse levels of learners for the present evaluation, this perspective validates that future iterations of this space should explicitly calibrate for different language levels. Finally, the time and level of action required was revealed to have contributed to some difficulty in the ability of 2 participants to navigate the experience; P11 revealed feeling overwhelmed with the dual actions they had to take at times: “It was difficult to imagine a sentence because it was difficult to take the wooden part. For the brain, it was too much (P11).” Another perspective was related to age, as P5 revealed their preference for experiences that did not require high levels of activity: “I think maybe for my age, I’m not very familiar with this kind of game, so I prefer to not have so much action, just like picking words, and go back to the firework and chatting with my friend (P5).” These insights reveal that **future design of virtual affordances can further emphasize reducing cognitive load for players, and consider time spent in the experience so that users do not experience physical discomforts.**

6 Limitations and Future Work

This initial design and evaluation revealed a number of valuable insights that will inform future refinements of the experience for broader evaluation. It would be valuable to evaluate technology comfort levels in order to account for factors such as novelty effect, which may have influenced the experiences and engagement level. Furthermore, we will explore through longitudinal studies with a more robust set of users the long-term impact of having access to this experience. Finally, we will conduct a comparative experimental study to evaluate self-efficacy measures before and following the virtual experience to understand how these affective experiences translate to emotional states outside the Virtual Reality setting.

References

- [1] Adinawa Adjagbodjou and Geoff Kaufman. 2024. Envisioning Support-Centered Technologies for Language Practice and Use: Needs and Design Opportunities for Immigrant English Language Learners (ELLs). In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–15. doi:10.1145/3613904.3642236
- [2] Miguel Barrera-Ángeles and Tilo Hartmann. 2022. Psychological benefits of using social virtual reality platforms during the covid-19 pandemic: The role of social and spatial presence. *Computers in Human Behavior* 127 (Feb. 2022), 107047. doi:10.1016/j.chb.2021.107047
- [3] Imogen H. Bell, Jennifer Nicholas, Mario Alvarez-Jimenez, Andrew Thompson, and Lucia Valmaggia. 2020. Virtual reality as a clinical tool in mental health research and practice. *Dialogues in Clinical Neuroscience* 22, 2 (June 2020), 169–177. doi:10.31887/DCNS.2020.22.2/valmaggia
- [4] Elin A. Björling, Rachael Cicero, Aditya Sankar, and Anand Sekar. 2019. Thought Disposal: Co-Designing a virtual interaction to reduce stress in teens. In *Proceedings of the 18th ACM International Conference on Interaction Design and Children (IDC '19)*. Association for Computing Machinery, New York, NY, USA, 562–567. doi:10.1145/3311927.3325313
- [5] Lisa S. Blackwell, Kali H. Trzesniewski, and Carol Sorich Dweck. 2007. Implicit Theories of Intelligence Predict Achievement Across an Adolescent Transition: A Longitudinal Study and an Intervention. *Child Development* 78, 1 (2007), 246–263. doi:10.1111/j.1467-8624.2007.00995.x _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-8624.2007.00995.x
- [6] Vassiliki Bouki, Daphne Economou, and Anastassia Angelopoulou. 2011. Cognitive theory of multimedia learning and learning videos design: the redundancy principle. In *Proceedings of the 29th ACM international conference on Design of communication (SIGDOC '11)*. Association for Computing Machinery, New York, NY, USA, 271–278. doi:10.1145/2038476.2038531
- [7] Pablo Briñol, Richard E. Petty, and Jennifer Belding. 2017. Objectification of people and thoughts: An attitude change perspective. *British Journal of Social Psychology* 56, 2 (2017), 233–249. doi:10.1111/bjso.12183 _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/bjso.12183
- [8] Deana Brown, Victoria Ayo, and Rebecca E. Grinter. 2014. Reflection through design: immigrant women's self-reflection on managing health and wellness. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (Toronto, Ontario, Canada) (CHI '14)*. Association for Computing Machinery, New York, NY, USA, 1605–1614. doi:10.1145/2556288.2557119
- [9] Alan Cheng, Lei Yang, and Erik Andersen. 2017. Teaching Language and Culture with a Virtual Reality Game. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. Association for Computing Machinery, New York, NY, USA, 541–549. doi:10.1145/3025453.3025857
- [10] Alice Chirico, Francesco Ferrise, Lorenzo Cordella, and Andrea Gaggioli. 2018. Designing Awe in Virtual Reality: An Experimental Study. *Frontiers in Psychology* 8 (Jan. 2018), 2351. doi:10.3389/fpsyg.2017.02351
- [11] Luca Cian. 2017. Verticality and Conceptual Metaphors: A Systematic Review. *Journal of the Association for Consumer Research* 2, 4 (Oct. 2017), 444–459. doi:10.1086/694082 Publisher: The University of Chicago Press.
- [12] Ruth Colvin Clark, author. 2016. *Applying the Personalization and Embodiment Principles: Use Conversational Style, Polite Wording, Human Voice, and Virtual Coaches*. Wiley., Hoboken .: doi:10.1002/9781119239086 Journal Abbreviation: E-learning and the science of instruction : Publication Title: E-learning and the science of instruction .:
- [13] Carol S. Dweck and Ellen L. Leggett. 1988. A social-cognitive approach to motivation and personality. *Psychological Review* 95, 2 (1988), 256–273. doi:10.1037/0033-295X.95.2.256 Place: US Publisher: American Psychological Association.
- [14] Dylan Ebert, Sanika Gupta, and Fillia Makedon. 2016. Ogma: A Virtual Reality Language Acquisition System. In *Proceedings of the 9th ACM International Conference on Pervasive Technologies Related to Assistive Environments (PETRA '16)*. Association for Computing Machinery, New York, NY, USA, 1–5. doi:10.1145/2910674.2910681
- [15] Jason B. Ellis, Kurt Luther, Katherine Bessiere, and Wendy A. Kellogg. 2008. Games for virtual team building. In *Proceedings of the 7th ACM conference on Designing interactive systems (DIS '08)*. Association for Computing Machinery, New York, NY, USA, 295–304. doi:10.1145/1394445.1394477
- [16] Gisela Ernst. 1994. "Talking Circle": Conversation and Negotiation in the ESL Classroom. *TESOL Quarterly* 28, 2 (1994), 293–322. doi:10.2307/3587435 Publisher: [Wiley, Teachers of English to Speakers of Other Languages, Inc. (TESOL)].
- [17] Sarah Garcia, Denis Laesker, Derek Caprio, Ronald Kauer, Jason Nguyen, and Marvin Andujar. 2019. An Immersive Virtual Reality Experience for Learning Spanish. 151–161. doi:10.1007/978-3-030-21817-1_12
- [18] Florian Grieger, Holger Klapperich, and Marc Hassenzahl. 2021. Trash It, Punch It, Burn It – Using Virtual Reality to Support Coping with Negative Thoughts. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems (CHI EA '21)*. Association for Computing Machinery, New York, NY, USA, 1–6. doi:10.1145/3411763.3451738
- [19] Alice Gruber, Silvia Canto, and Kristi Jauregi-Ondarra. 2023. Exploring the use of social virtual reality for virtual exchange. *ReCALL* 35, 3 (Sept. 2023), 258–273. doi:10.1017/S0958344023000125
- [20] Ying-yi Hong, Chi-yue Chiu, Carol S. Dweck, Derrick M.-S. Lin, and Wendy Wan. 1999. Implicit theories, attributions, and coping: A meaning system approach. *Journal of Personality and Social Psychology* 77, 3 (1999), 588–599. doi:10.1037/0022-3514.77.3.588 Place: US Publisher: American Psychological Association.
- [21] Crescent Jicol, Chun Hin Wan, Benjamin Doling, Caitlin H Illingworth, Jinha Yoon, Charlotte Headey, Christof Lutteroth, Michael J Proulx, Karin Petrini, and Eamonn O'Neill. 2021. Effects of Emotion and Agency on Presence in Virtual Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3411764.3445588
- [22] Keith Kenyon, Vitalia Kinakh, and Jacqui Harrison. 2023. Social virtual reality helps to reduce feelings of loneliness and social anxiety during the Covid-19 pandemic. *Scientific Reports* 13, 1 (Nov. 2023), 19282. doi:10.1038/s41598-023-46494-1 Number: 1 Publisher: Nature Publishing Group.
- [23] Alexandra Kitson, Ekaterina R. Stepanova, Ivan A. Aguilar, Natasha Wainwright, and Bernhard E. Riecke. 2020. Designing Mind(set) and Setting for Profound Emotional Experiences in Virtual Reality. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference (Eindhoven, Netherlands) (DIS '20)*. Association for Computing Machinery, New York, NY, USA, 655–668. doi:10.1145/3357236.3395560
- [24] Ming Kuo, Michael Barnes, and Catherine Jordan. 2019. Do Experiences With Nature Promote Learning? Converging Evidence of a Cause-and-Effect Relationship. *Frontiers in Psychology* 10 (Feb. 2019), 305. doi:10.3389/fpsyg.2019.00305
- [25] Haewoon Kwak. 2022. You Have Earned a Trophy: Characterize In-Game Achievements and Their Completions. In *Proceedings of the 14th ACM Web Science Conference 2022 (WebSci '22)*. Association for Computing Machinery, New York, NY, USA, 129–138. doi:10.1145/3501247.3531582
- [26] Yu-Ju Lan. 2015. Contextual EFL learning in a 3D virtual environment. (2015).
- [27] Victoria K. Lew, Joshua Smith, Nicholas Reuss, Xiyu Zhang, Jacob Gulliuzo, and Christina M. Frederick. 2019. Relatedness in gaming: the effect in-game touch has on player connection. *SN Applied Sciences* 2, 1 (Dec. 2019), 59. doi:10.1007/s42452-019-1859-4
- [28] Ping Li, Jennifer Legault, Alexander Klippel, Jiayan Zhao, et al. 2020. Virtual reality for student learning: Understanding individual differences. *Human Behaviour and Brain* 1, 1 (2020), 28–36.
- [29] Meei-Ling Liaw. 2019. EFL learners' intercultural communication in an open social virtual environment. *Journal of Educational Technology & Society* 22, 2 (2019), 38–55.
- [30] Jessica Lingel, Mor Naaman, and danah m. boyd. 2014. City, self, network: transnational migrants and online identity work. In *Proceedings of the 17th ACM Conference on Computer Supported Cooperative Work & Social Computing (Baltimore, Maryland, USA) (CSCW '14)*. Association for Computing Machinery, New York, NY, USA, 1502–1510. doi:10.1145/2531602.2531693
- [31] Pinyao Liu, Ekaterina R. Stepanova, Alexandra Kitson, Thecla Schiphorst, and Bernhard E. Riecke. 2022. Virtual Transcendent Dream: Empowering People through Embodied Flying in Virtual Reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–18. doi:10.1145/3491102.3517677
- [32] Peter MacIntyre and Tammy Gregersen. 2012. Emotions that facilitate language learning: The positive-broadening power of the imagination. *Studies in Second Language Learning and Teaching* 2, 2 (Oct. 2012), 193–213. doi:10.14746/ssllt.2012.2.2.4 Number: 2.
- [33] J Mak. 2016. Google Expeditions offers stunning field trips without leaving school. The International Society for Technology in Education.
- [34] Péter Kristóf Makai. 2021. Media and modalities—Computer games. In *Intermedial Studies*. Routledge, 69–85.
- [35] Richard E. Mayer. 2024. The Past, Present, and Future of the Cognitive Theory of Multimedia Learning. *Educational Psychology Review* 36, 1 (Jan. 2024), 8. doi:10.1007/s10648-023-09842-1
- [36] David B. Miele and Daniel C. Molden. 2010. Naive theories of intelligence and the role of processing fluency in perceived comprehension. *Journal of Experimental Psychology: General* 139, 3 (2010), 535–557. doi:10.1037/a0019745 Place: US Publisher: American Psychological Association.
- [37] David B. Miele, Lisa K. Son, and Janet Metcalfe. 2013. Children's naive theories of intelligence influence their metacognitive judgments. *Child Development* 84, 6 (2013), 1879–1886. doi:10.1111/cdev.12101 Place: United Kingdom Publisher: Wiley-Blackwell Publishing Ltd..
- [38] Brian Nelson and Younsu Kim. 2020. Multimedia design principles in game-based learning. In *Handbook of game-based learning*. The MIT Press, Cambridge, MA, US, 307–327.
- [39] Amy Oberfoell and A Correia. 2016. Understanding the role of the modality principle in multimedia learning environments. *Journal of Computer Assisted Learning* 32, 6 (2016), 607–617.
- [40] Seungoh Paek, Daniel L. Hoffman, and Antonios Saravanos. 2017. Spatial contiguity and incidental learning in multimedia environments. *British Journal of*

- Educational Technology* 48, 6 (2017), 1390–1401. doi:10.1111/bjet.12488 _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/bjet.12488>.
- [41] Paul K. Piff, Pia Dietze, Matthew Feinberg, Daniel M. Stancato, and Dacher Keltner. 2015. Awe, the small self, and prosocial behavior. *Journal of Personality and Social Psychology* 108, 6 (2015), 883–899. doi:10.1037/pspi0000018
 - [42] Andrew K. Przybylski, C. Scott Rigby, and Richard M. Ryan. 2010. A Motivational Model of Video Game Engagement. *Review of General Psychology* 14, 2 (June 2010), 154–166. doi:10.1037/a0019440 Publisher: SAGE Publications Inc.
 - [43] Richard W. Robins and Jennifer L. Pals. 2002. Implicit self-theories in the academic domain: Implications for goal orientation, attributions, affect, and self-esteem change. *Self and Identity* 1, 4 (2002), 313–336. doi:10.1080/15298860290106805 Place: United Kingdom Publisher: Taylor & Francis.
 - [44] Rini Kartika Sari and Andi Pratama. 2024. Integrating Gamification in Multicultural Classrooms A Study on Enhancing Collaborative Learning and Cultural Empathy. *International Journal of Educational Insights and Innovations* 1, 2 (Dec. 2024), 1–4. <https://ijedins.technolabs.co.id/index.php/ijedins/article/view/8> Number: 2.
 - [45] Yeşim Sevinç and Jean-Marc Dewaele. 2018. Heritage language anxiety and majority language anxiety among Turkish immigrants in the Netherlands. *International Journal of Bilingualism* 22, 2 (April 2018), 159–179. doi:10.1177/1367006916661635 Publisher: SAGE Publications Ltd.
 - [46] Kang Shumin. 2002. Factors to Consider: Developing Adult EFL Students' Speaking Abilities. In *Methodology in Language Teaching: An Anthology of Current Practice*, Jack C. Richards and Willy A. Renandya (Eds.). Cambridge University Press, Cambridge, 204–211. doi:10.1017/CBO9780511667190.028
 - [47] Chaklam Silpasuwanchai and Xiangshi Ren. 2018. A Quick Look at Game Engagement Theories. In *The Wiley Handbook of Human Computer Interaction*. John Wiley & Sons, Ltd, 657–679. doi:10.1002/9781118976005.ch29 Section: 29 _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781118976005.ch29>.
 - [48] Anselm Strauss and Juliet Corbin. 1998. Basics of qualitative research techniques. (1998).
 - [49] Robert T. Teranishi, Carola Suárez-Orozco, and Marcelo Suárez-Orozco. 2011. Immigrants in community colleges. *The Future of Children* 21, 1 (2011), 153–169. doi:10.1353/foc.2011.0009
 - [50] Tricia Thrasher. 2022. The Impact of Virtual Reality on L2 French Learners' Language Anxiety and Oral Comprehensibility: An Exploratory Study. *CALICO Journal* 39, 2 (June 2022), 219–238. doi:10.1558/cj.42198
 - [51] Melanie Trypke, Ferdinand Stebner, and Joachim Wirth. 2023. Two types of redundancy in multimedia learning: a literature review. *Frontiers in Psychology* 14 (May 2023), 1148035. doi:10.3389/fpsyg.2023.1148035
 - [52] Vincent van Brakel, Miguel Barrada-Ángeles, and Tilo Hartmann. 2023. Feelings of presence and perceived social support in social virtual reality platforms. *Computers in Human Behavior* 139 (Feb. 2023), 107523. doi:10.1016/j.chb.2022.107523
 - [53] Matija Vigato and Tihana Babić. 2021. Research on Gamer Motivation Factors Based on the Gamer Motivation Model Framework. In *2021 44th International Convention on Information, Communication and Electronic Technology (MIPRO)*. 730–735. doi:10.23919/MIPRO52101.2021.9596942 ISSN: 2623-8764.
 - [54] Gregory M. Walton and Alia J. Crum (Eds.). 2021. *Handbook of wise interventions: How social psychology can help people change*. The Guilford Press, New York, NY, US. Pages: xiii, 474.
 - [55] Zhiguo Wang and Hongwei Na. 2023. Multimedia Technology Based Interactive Translation Learning for Students. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* (March 2023). doi:10.1145/3588569 Just Accepted.